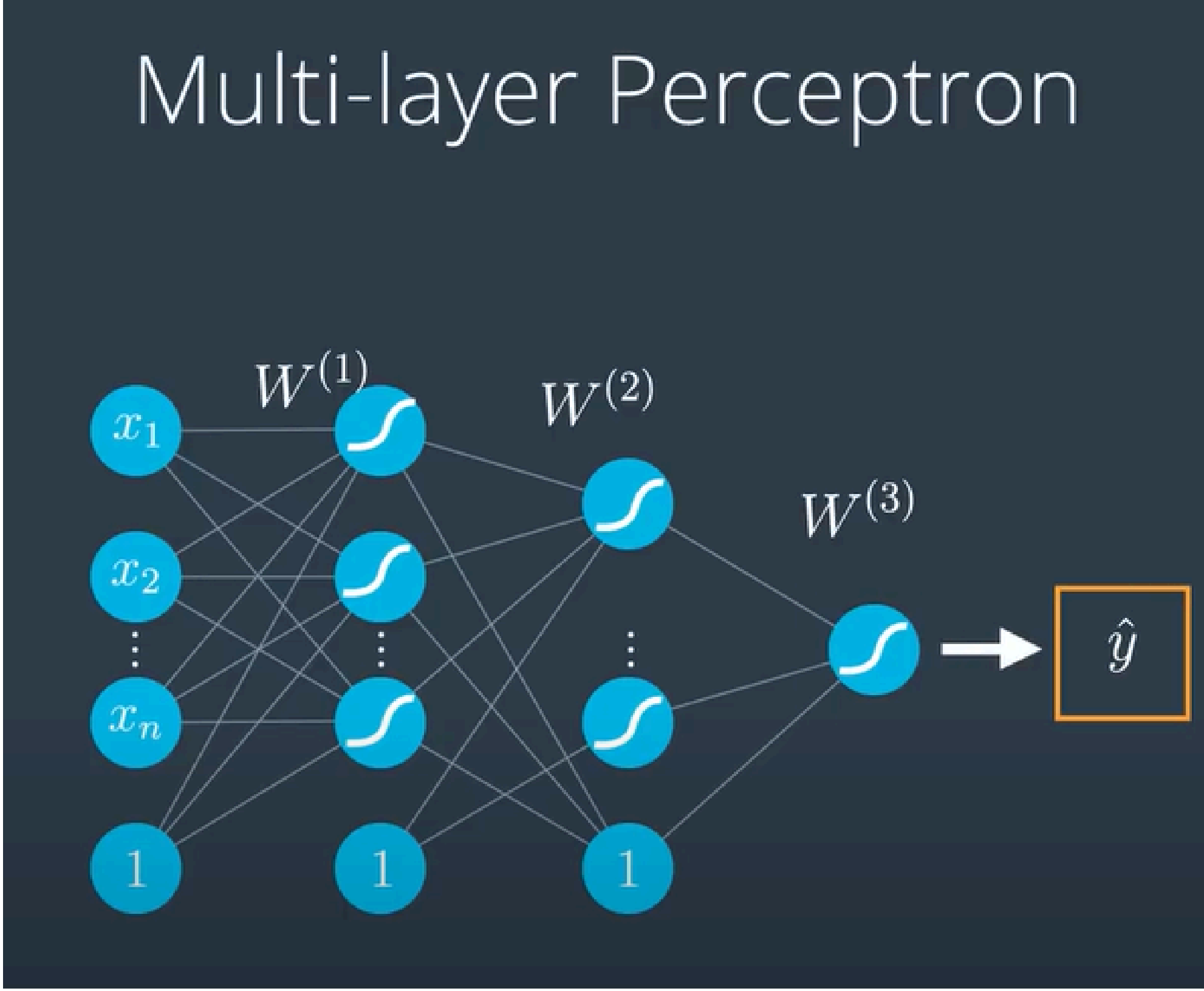


Neural Network Architecture

Feedforward



PREDICTION

$$\hat{y} = \sigma \circ W^{(3)} \circ \sigma \circ W^{(2)} \circ \sigma \circ W^{(1)}(x)$$

ERROR

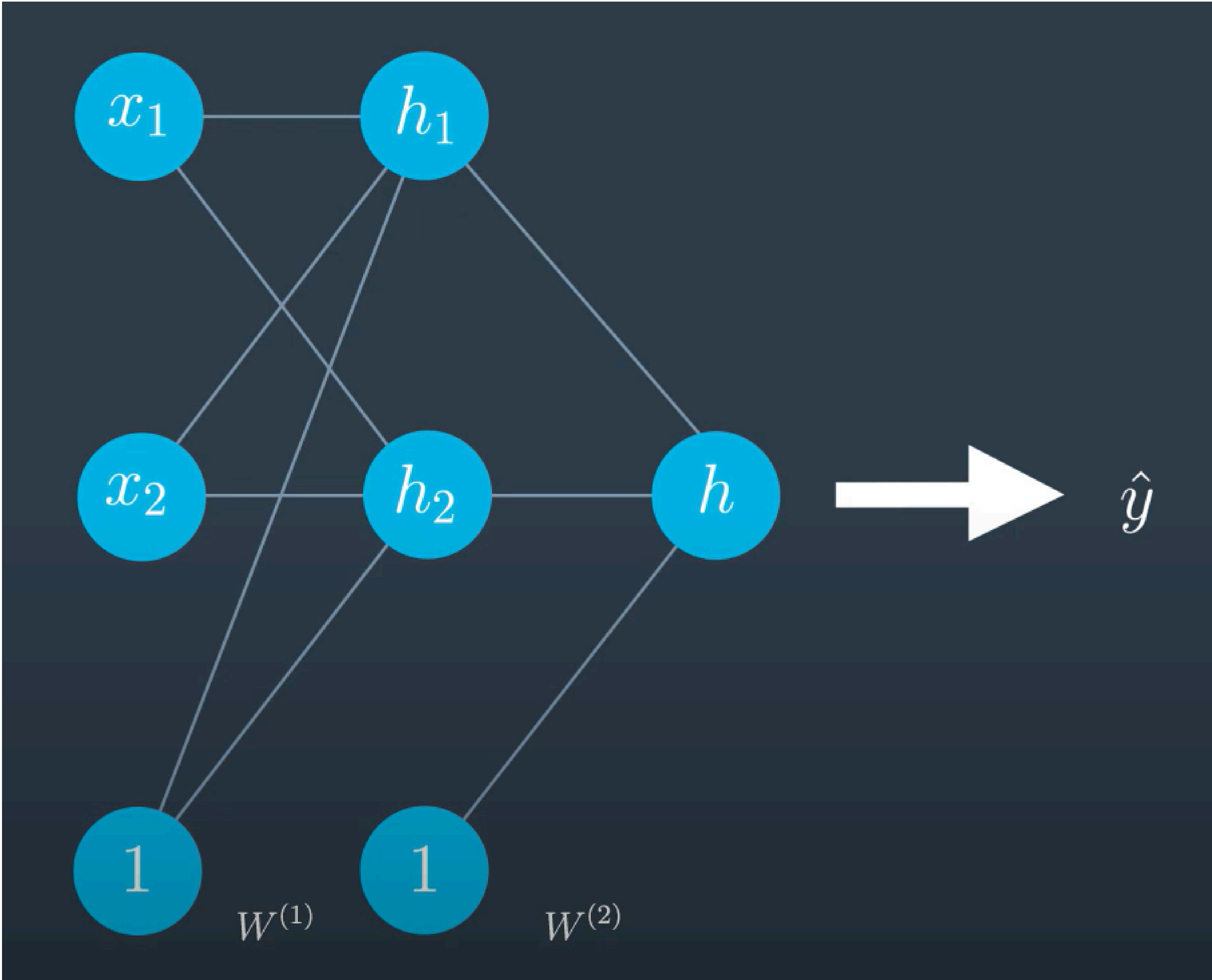
$$E(W) = -\frac{1}{m} \sum_{i=1}^m y_i \ln(\hat{y}_i) + (1 - y_i) \ln(1 - \hat{y}_i)$$

Backpropagation

Now, we're ready to get our hands into training a neural network. For this, we'll use the method known as backpropagation. In a nutshell, backpropagation will consist of:

- Doing a feedforward operation.
- Comparing the output of the model with the desired output.
- Calculating the error.
- Running the feedforward operation backwards (backpropagation) to spread the error to each of the weights.
- Use this to update the weights, and get a better model.
- Continue this until we have a model that is good.

Sounds more complicated than what it actually is.

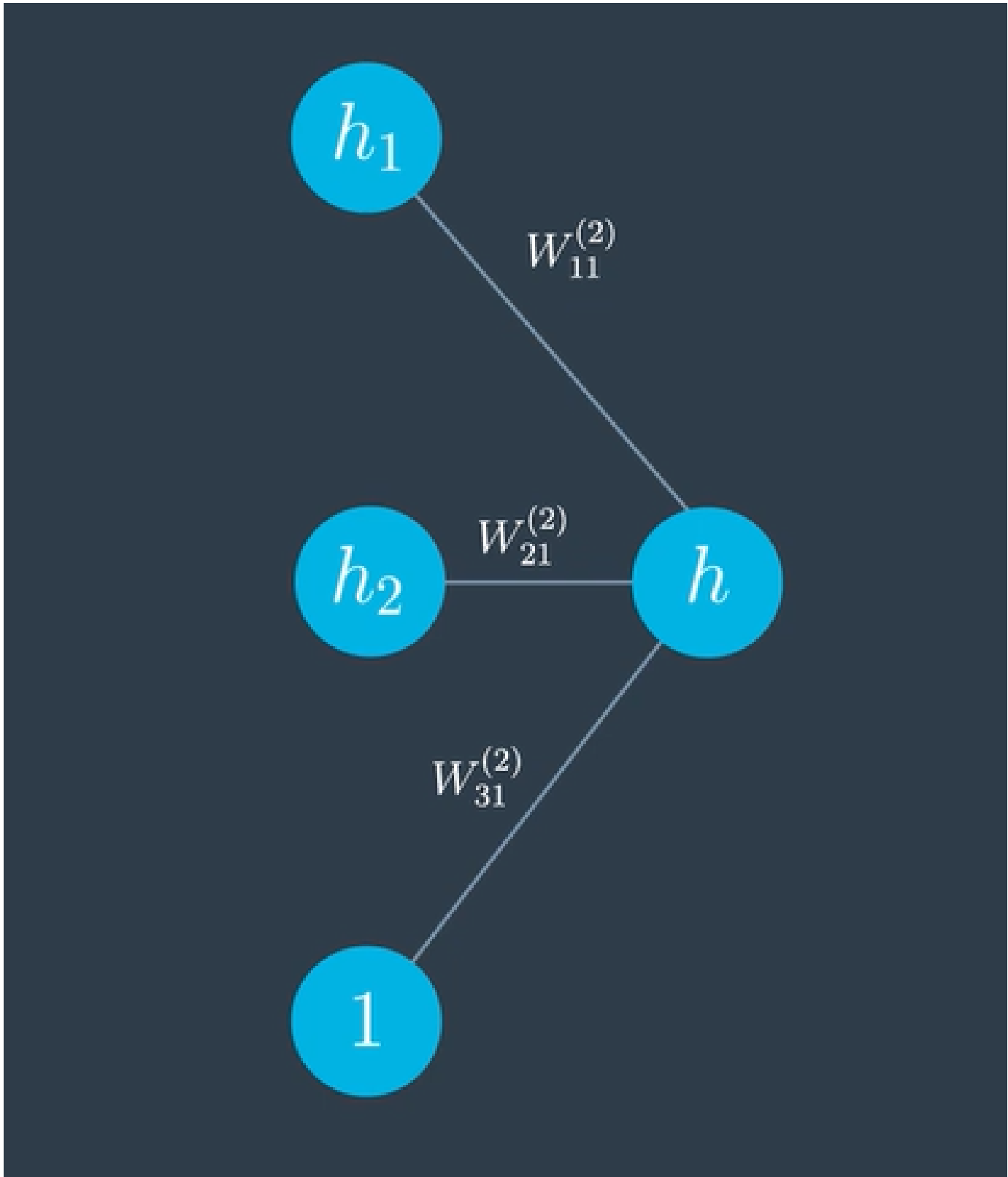


$$E(W) = -\frac{1}{m} \sum_{i=1}^m y_i \ln(\hat{y}_i) + (1 - y_i) \ln(1 - \hat{y}_i)$$

$$E(W) = E(W_{11}^{(1)}, W_{12}^{(1)}, \dots, W_{31}^{(2)})$$

$$\nabla E = \left(\frac{\partial E}{\partial W_{11}^{(1)}}, \dots, \frac{\partial E}{\partial W_{31}^{(2)}} \right)$$

$$\frac{\partial E}{\partial W_{11}^{(1)}} = \frac{\partial E}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h} \frac{\partial h}{\partial h_1} \frac{\partial h_1}{\partial W_{11}^{(1)}}$$



$$h = W_{11}^{(2)} \sigma(h_1) + W_{21}^{(2)} \sigma(h_2) + W_{31}^{(2)}$$

$$\frac{\partial h}{\partial h_1} = W_{11}^{(2)} \sigma(h_1) [1 - \sigma(h_1)]$$